

METHODOLOGY — draft 2026-08-28

Status: complete-details draft for the author to rewrite. Every concrete value carries a pointer (file:line in the repo at HEAD 8b29524, branch 4dof-cartesian; artifact paths under the off-cluster mirror ~/workspace/final_rr_artifacts_2026-08-24/ = M/; cluster-only source under /cluster/tufts/shortlab/jstale02/ = C/). Where sources disagree both values are given and flagged **[CHECK]**; the full **[CHECK]** list is collected in §8. Precedence when documents conflict: paper/AUDIT_results_2026-08-28.md > paper/PREREG_final_round_robin_2026-08-23.md (+A1-A10) > code > older narrative docs. Result numbers are deliberately *not* reproduced here except where they characterise an instrument (fidelity, survivors, ceilings); results live in paper/RESULTS_TABLE_2026-08-25.md (regenerate with analysis/results_table.py).

Notation: d{source}_{learner}[-dense]_s{seed}; sources dH (human), dDP (diffusion-policy teacher), dR2D (r2dreamer teacher); learners DP, RLPD, r2dreamer (r2d), dv3.

LIVING UPDATES (2026-08-29, kept current by the agent during the cluster block).

Sections carry an

UPDATE 08-29 block wherever the recipe, seeds, or rules changed after this draft. Precedence: PREREG amendments

A16-A19 > these blocks > the 08-28 text. Stable material the user is writing lives outside this file.

1. Real robot and demonstration data

1.1 Platform and teleoperation

item	value	pointer
Robot	Kinova gen3-lite , 6 revolute arm joints (joint_1..6) + 2-finger gen3-lite gripper (kortex_description/grippers/gen3_lite_2f); ROS namespace /my_gen3_lite	trial_read.py:8-9, 23-25; kinova.py:1-8; URDF gen3_lite_2f_robotiq

	(PITCH_CAP, = measured max \		
4- DOF verifi cation	recorded wx=wz=0 in 25/25 inspected demos, wy≠0 in 25/25; over all 96 *_cartesian. npz tapes per-dim max \	v\	= [0.11, 0.11, 0.11, 0, 1.0, 0]; the plugin's Jacobian n control ler left roll/yaw free (drift up to 1.03/1. 34 rad)
Grip per sign al	base_feedbac k... gripper_feed back[0].moto r[0].positio n, 0 open ... 100 closed (observed range −0.98... 100.69 in tapes); "closed" threshold GP_CLOSE = 30	trial _read er.py :14; examp le.py :206- 207; can_p os_re cover y/rep lay_h arnes s.py: 60	cartesia n_env.py :80-84; measur ed from inthewil d_trials /*_carte sian.npz

1.2 What was recorded (ROS bags → inthewild_trials/)

item	value	pointer
Topics (original reader)	/my_gen3_lite/joint_states (position[:6]), /joy (read, never stored), /my_gen3_lite/base_feedback	trial_reader. py:1-2,11- 26,30,50-51

	(tool_pose x,y,z,θx,θy,θz; gripper motor position)	
Topics (cartesian reader)	additionally <pre>/my_gen3_lite/in/cartesian_velocity (TwistCommand) and base_feedback.tool_twist_*; /reward,/success deliberately not extracted (untrusted auto-detector)</pre>	trial_reader_cartesian.py:17,30-32,47-56
Windowing	frame emitted whenever $\geq 1/60$ s elapsed and both topics have ≥ 1 msg in the window; values are per-window means; last frame dropped	trial_reader.py:32,43-54
Effective rate	nominal "30 Hz" (DATA_HZ = 30 # approx -- timestamps weren't saved; user_232 was ~31.6 Hz (919 frames / 29.06 s)); joint_states itself is ~40 Hz (dt p95 25.4 ms); *_cartesian.npy for uid 232 has 1159 frames (~39.9 Hz) [CHECK 1]	example.py:276; CAN_STARTING_POSITION.md:188; cartesian_env.py:51 (DT = 0.025)
Output (<uid>_episodes.npy)	pickled dict {'vel_cmd': (T,6) float64, 'gripper_pos': (T,1)}. NB: despite the key name, vel_cmd is the measured joint position stream (/joint_states position[:6]), replayed as PD position targets; several docs call it "commanded joint targets" [CHECK 2]	trial_reader.py:69-74; example.py:284; real2sim_follower_lab_2026-08-23.md:85-86; trial_reader_cartesian.py:11
Superset (<uid>_cartesian.npy)	<pre>joint_pos, vel_cmd(=joint_pos), tool_pose (m; θ in degrees), tool_twist, gripper_pos, cartesian_velocity, cartvel_reference_frame float32; tool_pose is in the ROBOT BASE frame; sim world z = base z + 0.05</pre>	trial_reader_cartesian.py:10-16,79-80; can_pos_recovery/read_bag.py:7
Cameras	per-trial MP4s (raw/user_<id>/cam_dev_video0 bottom-up through the	CAN_STARTING_POSITION.md:123-124,189-190,218,283-

	translucent shelf, <code>cam_dev_video4</code> top view of the start), stored rotated $\sim 90^\circ$, no factory calibration; resolution/model not documented anywhere [CHECK 3] ; cameras are NOT in the bag reader and NOT used by any learner (sim cameras only, §3.2)	284; HANDOFF_PLAN. md:80-82
Object pose	never recorded (motivates §2.3)	CAN_STARTING_ POSITION.md:8 -9
Require ments	original reader needs ROS1 <code>rosvbag</code> ; cartesian reader uses pure-python <code>rosvbags</code> ; raw bags are not on the devbox	<code>trial_reader.</code> <code>py:1</code> ; <code>trial_reader</code> <code>cartesian.py:</code> 5-7,38

1.3 Trial census and labels

count	meaning	pointer
96	<code>*_episodes.npy</code> files on disk, uids 232-335 (missing 241,264,271,272,291,292,32 3,332); = union of the label lists in <code>kinova.py</code>	<code>ls</code> <code>inthewild</code> <code>trials</code> ; <code>trial_re</code> <code>ader.py:</code> 78-86; <code>kinova.p</code> <code>y:16-21</code>
93	trials in <code>can_pos_recovery/trial_placeme</code> <code>nts.json</code> and <code>baselines/demo_manifest.json</code> (77 success / 16 fail). uids 253, 285, 289 (fail-list) are on disk but absent from both, reason undocumented [CHECK 4] . Docs say "93 in- the-wild trials recorded"	PAPER_PL AN.md:75 -80; paper/ME THODOLOG Y.md:65- 72
2 stubs	290, 322: gripper never closes; excluded	CLAUDE.m d; CAN_STAR TING_POS ITION.md

		:56,68,72,214	
91	graded = baselines/demo_manifest_auth.json (75 success / 16 fail)	measured	
16 fail-labeled	238 240 249 260 268 270 282 288 296 307 310 312 313 314 324 334 (label bug: 322 and 331 appear in both a success and a fail list in kinova.py)	demo_manifest.js on; CAN_STARTING_POSITION.md:154,211	
9	success-labeled but never pick under replay: 236 237 239 244 255 261 294 303 331	demo_manifest_auth.json	
66	IL-usable human demos = 75 – 9; by furthest replay stage: picked-only 2 / placed 38 / contact 6 / nested 20	PAPER_PLAN.md:60-71,75-80; crosstab of demo_manifest_auth.json	
Frames	66 IL tapes = 132,365 frames; all 96 tapes min 356 / median 1530 / max 6233	measured from n fields	
Rule	IL learners never train on failed demos (user rule 2026-07-30); fails are RL/WM negative data only	PAPER_PLAN.md:357; METHODOLOGY.md:154-156	
`-p 0	1	2`	can-position bucket filter in example.py (nomi

nal
mark
ed
circle
s
POSITI
ON_0=(
0.4381
,0.1),
_1=(0.
4381,-
0.05),
_2=(0.
4381,-
0.2);
bucke
ts
wron
g by
up to
~5
cm,
super
seded
by
per-
trial
recov
ery
§2.3);
bucke
t
count
s
38/27
/28

Unla
bele
d
200-
231

never ingested ("~130
recordings" claim
inconsistent with a 32-wide
range **[CHECK 5]**)

HANDOFF_
PLAN.md:
91-93

1.4 Task and geometry

ite m	value	pointer
Task	pick a soup can from the table	paper/

	region and place/nest it against a static goal can on a shelf; stages picked → placed → contact → nested; the study is scoped to the pick phase	METHODOLOGY.md:4 5-48; PREREG §0
Can (real)	Campbell's soup can 66 × 101 mm, ≈0.35 kg	CAN_STARTING_POSITION.md:170-171
Can (sim)	cylinder r 0.033, h 0.101, p 1000 → 0.345 kg; can_friction 0.2 (inert: finger-can pair takes max(1.0,0.2))	trial_placements.json['world']; replay_harness.py:125-128; gripper_lab_2026-08-25.md:62-64
Goal can	same cylinder, p 1000, friction 2.0, static at STATIC_BOTTLE_POSITION = (0.672, -0.221, 0.19)	replay_harness.py:51,129-132
Goal recovery	user-validated demos 233 + 242 ("pick/place/slide perfect") must end touching the goal → two-circle intersection, south candidate chosen (matches slide directions). Rejected alternatives: fixed-radius ring fit on 61/75 real tool_pose slides → (0.662, -0.057) (rms 25 mm; biased by one-sided north approaches); placement-cluster median (0.656, -0.103); ancient (0.6, -0.2)	replay_harness.py:43-51; can_pos_recovery/goal_from_slide_s.py:1-19, goal_from_slides.json; CAN_STARTING_POSITION.md:450-476; CLAUDE.md 07-20
Real shelf	open acrylic rack; real shelf height above the table is unmeasured (decides shelf6 vs shelf10; inert for pick scope, blocks place/nested claims)	CAN_STARTING_POSITION.md:190,208-210; paper/AUDIT_INDEX.md:82-83; UPDATE_2026-08-25.md:157-158
Start pose	fixed arm start (HARDCODED_START), real tool pose at start ≈ (0.367, 0.011, 0.090) base frame, ~constant across demos	example.py:180; replay_harness.py:62-64; cartesian_env.py:54
Pred	see §3.5 (pick) and §6.2	replay_harness.py

icat es	(placed/contact/nested; NESTED_TOUCH_DIST = 0.066+0.015 = 0.081 m)	y:52-56
------------	--	---------

2. Real-to-sim

2.1 Simulator pin

item	value	pointer
Engine	Genesis, upstream github.com/Genesis-Embodied-AI/Genesis @ 31951c3f (reports 0.2.1; $\approx 0.2.1$ + ~270 upstream commits) + 2-file headless-render patch; reproduce: clone, checkout 31951c3f, git apply cluster/patches/genesis_0.2.1_headless_render.patch, pip install -e .	cluster/patches/GENESIS_PIN.md:1-16; commit 37b8d8f; gripper_lab_2026-08-25.md:30-31
Patch content	genesis/ext/pyrender/renderer.py: per-env camera poses (env_poses) so one batched render serves N wrist cameras; genesis/vis/camera.py: use_imshow=False gate on all cv2.imshow/waitKey (headless)	cluster/patches/genesis_0.2.1_headless_render.patch
Engine gate	Genesis 1.2.1 rejected: 3.4x lower contact force, pinch grasp fails	CAN_STARTING_POSITION.md:391-418
Software	python 3.10; torch 2.7.0+cu126; taichi 1.7.4; numpy 2.2.6; cv2 4.11; one pip-only conda env serves genesis + lerobot + SB3 (conda-forge libstdc++ poisoned the stack 07-30 → "NEVER conda-install"); r2dreamer in a separate py3.11 venv (torch 2.8.0+cu126)	CLAUDE.md:7; cluster/install_lerobot.sh:7-10,44-63; cluster_dv3_setup.sh:16; cluster/install_r2dreamer.sh:8,30-31; cluster/verify_env.sh:3-8,24-58
Physics	gs.init(precision="32", seed=0);	replay_harness.py

	SimOptions(dt=0.01, substeps=8) → substep_dt 1.25 ms; 3 physics steps per env step (30 Hz env clock); CPU backend for all training/eval worlds (one world per process)	:96-97,107-109; genesis_can_env.p y:14,86,245-246; trial_placements. json['world']
Contact solver defaults	every geom carries MuJoCo- default sol_params (timeconst 0.02 s, dampratio 1, dmin 0.90, dmax 0.95, width 1e-3, mid 0.5, power 2); this tree only max()-clamps timeconst at 2·substep_dt = 0.0025 s (the old comment in replay_harness.py:98-99 claiming 0.2.1 hard-wires it is wrong for this tree) [CHECK 6]	gripper_lab_2026- 08-25.md:103- 104,115-121
Determinism	replay is machine- and load- dependent (cross-box placed +4/75; heavy CPU load flips borderline demos) → same- machine baselines, idle box, ×3	paper/ METHODOLOGY.md:58 -62; CLAUDE.md 07- 20 BB

2.2 Scene (can_pos_recovery/replay_harness.py: :build_world, lines 78-183)

object	spec	pointer
Table (table=True)	fixed box (0.419, 1.2, 0.05) at (0.3395, -0.1875, 0.025); top TABLE_TOP_Z = 0.05; friction 0.5	:76,114-121
Shelf ("box")	box BOX_SIZE=(0.4,0.75,0.12) at BOX_POS=(0.75,-0.1875,0.05) → BOX_TOP_Z = 0.11 (only 6 cm above the table in the base world); ρ 1000 (~36 kg), friction 0.5; dynamic in base, fixed=True under variants	:38-42,111- 113; sim_variants.p y:84-88,111- 122
Robot	gen3_lite_2f_robotiq_85.urdf at (0,0,0.05), fixed; arm kp [200,200,150,100,60,60], kv	:81,122- 124,171-179; sim_variants.p

	<pre>[20,20,15,10,6,6], force ±[50,50,50,20,20,20]; fingers kp 40 (finger_kp), kv 10, force ±50 (finger_force)</pre>	y:27-30
Gripper	<pre>4 finger DOFs with mimic (tip = -0.676·bottom + 0.149) enforced as equality constraints; reading 0..100 mapped linearly to driver $\theta \in [-0.09, 0.96]$ rad; collision = one convex hull per finger mesh</pre>	<pre>gen3_lite_2f_r obotiq_85.urdf :313; replay_harness .py:66-72; gripper_lab_20 26-08- 25.md:109- 110,127-134</pre>
Cans	§1.4	
Cameras (camera_rig=True)	<pre>two 64×64 RGB cams: top at (0.40,-0.08,1.15) looking down (up=(1,0,0), fov 68); wrist fov 80 attached to eef with offset (0.10, 0, -0.03), pitched 30°, rolled 90°; obs image = top ++ wrist = (64,64,6) u8. Video cam (not an obs) 640×560</pre>	<pre>replay_harness .py:141-170; genesis_can_en v.py:89,143- 149</pre>

2.3 Can starting-position and goal recovery

- Method: FK at the grasp instant (closure interval that lifts and releases near the goal; grasp = reach-bottom) → fingertip midpoint = can xy; validated against bag tool_pose to ~1 cm xy / ~1 mm z; fk_recovered.json covers all 96 trials (CAN_STARTING_POSITION.md:40-53,181-187; can_pos_recovery/recover_can_position.py, fk_all_trials.py).
- Placement search around the FK seed: search_placements.py, batch_harness.py (GPU B=32), cpu_research.py (CPU-parallel, ok-class by construction, ~25 s/rollout, 16-way) → trial_placements.json (CAN_STARTING_POSITION.md:192-203; cpu_research.py:1-17). Current file: 93 trials; 72 "solved" (ok_batch 52, ok 20); 20 no_shelf_batch, 1 shelf_but_no_slide; unsolved 255, 303 (CLAUDE.md 07-20 BB).
- Independent panel (2026-07-08) overturned a rigged negative control and a coverage overstatement (winner drift up to 17 cm; 60 % depended on goal relocation); fixes baked in: fail-labeled demos never receive relocated goals; nested requires picked (CAN_STARTING_POSITION.md:254-305; CLAUDE.md).
- World corrections found 2026-07-04: table added; can geometry/mass corrected (was 70×75 mm p 2000); finger gains restored; substeps 1→4→8 (pick/run 0.59→0.74, 16==8 converged) (CAN_STARTING_POSITION.md:160-178,341-349). Note doc says finger kp 100 "sweet spot" but the live world uses finger_kp 40 (commit e8a07d6, 07-14) **[CHECK 7]**.
- --ic-from-tape (PREREG A3): the 12 IL demos without a recovered placement

(uids 233 259 262 266 267 275 278 301 319 321 329 333) reset to the can pose at frame 0 of their own stride-1 sim tape (states[0][8:11], quat [11:15]), static goal; applied identically to all three sources so every source records from the same 66 ICs (baselines/record_demos.py:662,731-757; PREREG:248-250; SESSION_LOG_2026-08-23_cluster.md:21).

- Goal: §1.4. Whole-corpus replay at the final goal (75 success demos, base world): picked 61 / placed 47 / contact 15 / nested 16; fail-demo negative control nested 2/16 (CLAUDE.md 07-20; METHODOLOGY.md:99-102).

2.4 The human-command follower ("human" adapter)

The human demos are not re-encoded offline; they are **re-executed** inside the learners' MDP (§3) by a closed-loop waypoint follower, so every source is "recorded as executed" by one recorder.

item	value	pointer
Source tape	stride-1 sim tape from baselines/episodes_pick_phase_dpuned (the 66 IL demos, DP-pruned, §4.5); cmd = actions[:, :6] (real measured joints used as targets), ref = states[1:, :6] (old-sim arm path), grip = clip(actions[:,6])	record_demos.py:416-421,656; real2sim_follower_lab_2026-08-23.md:82-88
Per decision	target waypoint jt = min(j+3, n-1) (≤4 waypoints per decision); a_arm = clip((cmd[jt] - target_now)/(4·0.025), -1, 1); a_grip = grip[jt]·2 - 1	record_demos.py:439-444
Arrival rule	scan k from jt down to j; advance to k+1 if q_meas - ref_k ∞ < tol (meas) OR, with --arrival either, target_env - cmd_k ∞ < tol; tol = 0.025 rad ; max_dwell = 8 decisions without arrival → forced jump (counted as a stall); dilation_cap = 3.0 (step cap = [3·n/4] decisions → adapter_exhausted); settle = 25 decisions holding the last waypoint; dilation = decisions·4/n_src	record_demos.py:427-437,448-470,668-671; manifests M/demos_v1/dH_either/manifest.json configs[0] (tol 0.025, max_dwell 8, arrival either, dilation_cap 3.0, settle 25)
Rule	--arrival either (PREREG A6; lab:	PREREG:272-276;

le of re co rd	49→55 local survivors, zero dwell stalls, identical path fidelity; commit 33d5bc9). The block's dH sets (matched_v2, matched_w3) are either; the original 51-tape set (matched_v1) and the dDP teacher's training data are meas (disclosed, A6)	real2sim_follower_lab_2026-08-23.md:23-31,132-133,274-276; SESSION_LOG:74-75
Ke ep rul e	kept iff the hardened pick (§3.5) is re-earned; drops reported as a source property (A1)	PREREG:241-243

2.5 Sim variants (baselines/sim_variants.py)

Mechanics: `install(name)` monkeypatches `gs.Scene.add_entity` before build (robot gets `Rigid(gravity_compensation=gc)` and `pos.z += riser`; shelf gets `pos.z += shelf_dz`, `fixed=True`); `post_build(w, name)` sets arm `kp/kv = BASE × mult`, force range, optional `grasp_timeconst`, `goal_start_z += shelf_dz (:93-163)`. Plumbed via `baselines/sim_variant_hook.py` `apply_pre/apply_post`; 'base' = unpatched; stamped into every tape/manifest/sidecar/registry row; eval resolution "explicit CLI > checkpoint sidecar > 'base', mismatch = refuse" (`sim_variant_hook.py:1-10`; `record_demos.py:115-120,617-633,669`; `sbatch_dp.sh:195-196,300-303,475`; `sbatch_rlpd.sh:87,169-172,190,209`; `sbatch_r2dreamer.sh:209-212,252-258`; `wandb_eval.py:122-140`).

variant	kp / kvx	gravity comp	riser	shelfdz	grasp_timeconst	
base	1/1	0	0	0	— (engine default 0.02s)	: 3 5
gc_kp4_riser3	4/2	1.0	0.	0	—	: 5 1

gc_kp
4_ris
er3_s
helf6
(wor
ld of
reco
rd
for
the
corr
ecte
d
bloc
k)

4/
2

1.
0

0
.03

0
.06
(
s
h
e
l
f
t
o
p
0
.17
w
o
r
l
d
)

—

:
6
7

gc_kp
4_ris
er3_s
helf6
_ts5
(pilo
t
only
)

4/
2

1.
0

0
.03

0
.06

0.00
5 s
on
the
4
finge
r
geo
ms
AND
the
pick
ed
can
(Gen
esis

:
6
9
-
7
8
,
1
3
8
-
1
5
6

aver
ages
a
pair'
s
sol_p
arams
, so
both
sides
must
mov
e);
floor
tmin
=
2·sub
step_
dt
enfo
rced
by
max()

gc_kp
4_ris
er3_s
helf1
0
(dat
a-fit
alter
nati
ve)

4/
2

1.
0

0
.
0
3

0
.
1
0

—

:
8
0
-
8
2

othe
rs
(gc,
kp2,
kp4,
gc_kp
8,
*_urd
f,
riser
2/4,
fcap2
,
fmap)

a
bl
a
ti
o
n
la
d
d
e
r

:
3
6
-
6
0

kp×4 → arm kp [800,800,600,400,240,240], kv×2 (SESSION_LOG:136). post_build prints but does **not assert** the 5-geom count (CRITIQUE Q8) **[CHECK 8]**
(sim_variants.py:156; CRITIQUE_decisions_2026-08-26.md:364-368).

2.6 Physics defects found against real bag data, fixes, and fidelity

Source docs: paper/real2sim_follower_lab_2026-08-23.md (defects a-d) and paper/gripper_lab_2026-08-25.md (defect e). All numbers are local-devbox unless marked cluster.

defect	evidence	fix
(a) no arm gravity compensation + soft PD (sag τ_g/kp ; ~4 cm eef sag at the extended place pose, uid 242 sim eef 0.147 vs real 0.191)	real2sim... :166,171- 176,460- 462	gravity_comp 1.0, kp×4 / kv×2
(b) bandwidth: leash-limited steady speed ≈ 0.75 rad/s vs real p99 0.6–1.35 rad/s	:178-182	kp×4
(c) robot mounted ≥ 3 cm too low relative to the table (real tool z reaches $-0.019\dots+0.013$ base-frame; sim finger hulls bottom out at tool z $\approx +0.03$) → "pressing" failures 295/299/254/256/286/298/330 and the 246 knock-over	:184- 195,208- 215; sim_varia nts.py:46 -49	riser 0.03
(d) shelf modelled half-buried (top 6 cm above table; real release tool z 0.17-0.21 $\Rightarrow \geq 12$ cm) — masked in the base world because the sag lowered the can onto the low shelf (two cancelling errors: riser-only world nested 20→3)	:456-467; UPDATE_20 26-08- 25.md:91- 96; sim_varia nts.py:61 -66	shelf_dz 0.06 (shelf10 = data-fit upper bound; real height unmeasured)
(e) contact-solver timeconst 0.02 s = 8× the engine floor 2·substep_dt = 0.0025 s; penetration $\propto$ timeconst ² ; the old "finger reading map is miscalibrated" theory refuted	gripper_l ab:25- 35,46- 53,103- 121,184- 216	grasp_timecons t 0.005 (ts5) — pilot only, not adopted for any result

(real gripper over-travels into rubber/current limit at readings 0.667–1.007 on the same can)

block (\$2.7)

Fidelity (replaying the real robot's recorded joint stream through the follower, 66 tapes, 39 332 frames):

metric	base	gc_kp4_riser3 / shelf6	+ts5	pointer
joint tracking $\ e\ _\infty$ p50 / p95 (rad)	0.047 / 0.100 (whole-corpus figure quoted as 0.045 / 0.090)	0.004 / 0.030 (0.003 / 0.029 in the 75-demo table; 0.0033 / 0.0285 cluster)	0.0035 / 0.0300	real2s image: 162-170, 199-208, 441-451; UPD ATE: 87-89, 104-108
frames beyond the leash (0.125 rad)	7.9 % (4.0 % in the 75-demo table)	0.1-0.2 %	unchanged	same
frames beyond the cap	87 %	10 %	—	real2s image: 199-208
tracking error at grip close	0.049	0.002	—	same
pick re-earned	—	64/66	—	real2s

under open-loop replay					im... :41 -46
					rea l2s im... :44 1-
full-task 75-demo replay				69/6	451;
picked/pl aced/plac ed_v2/co ntact/nes ted/tippe d	67/59/2 6/18/20/ 32	71/55/28/22 /17/38	0/31 /20/ 20/2 3		gri ppe r_l ab: 521 - 527 ,62 3- 640
carry penetrati on med- of-max / p95	9.7–9.9 mm / 9.0 mm	9.9 mm	1.5 mm / 0.90 mm		gri ppe r_l ab: 623 - 640; UPD ATE :10 4- 108
					UPD ATE :10 6; CRI TIQ UE_ dec isi ons :37 -42
free tip- overs / 75	—	20	12 (10 reco verie s vs 2 losse s, McN ema r p≈0. 039)		
grip force med-of- max (obs[7]-	143 N	147 N	245 N (fing ers		gri ppe r_l ab: 54-

related)

stall
at
the
surfa
ce
holdi
ng
full
PD
error
;
ever
y
redu
ction
lever
loses
the
gras
p;
engi
ne
has
no
torsi
onal
fricti
on/i
mpr
atio)

61,
737
-
758

human
pick
recreation
(recorder,
/66)

51
(meas)
→ 56
(either)
old
world

57 base-
control vs
58 ts5
(cluster-
confirmed;
ts5 rescues
246, loses
316/254) —
but 2 vs 1
discordant,
McNemar
p=1.0

random-
teacher
control

0/30
cluster,
0/90
local

E5 caveat

ts5 pair-
averagi
ng also
stiffens
can↔go
al-can
and
can↔sh
elf
contacts
0.020→0
.0125 s
(1.6×)
— the
metric-
bearing
contacts
for
contact/

UPD
ATE
:10
0-
109;
gri
ppe
r_l
ab:
459
-
486;
CRI
TIQ
UE_
dec
isi
ons
:37
-42

UPD
ATE
:10
4;
gri
ppe
r_l
ab:
603
-
607

AUD
IT_
IND
EX.
md:
75-
77;
CRI
TIQ
UE_
dec
isi
ons
:18
2-
201

nested

[CHECK 9] the "57/66 vs 58/66 pick recreation" and "56/66 either" figures come from different runs (gripper-lab g_base control vs the 08-23 dH_either recording vs the 08-24 dH_w2 recording = 58/66 in gc_kp4_riser3_shelf6); state which one is meant.

2.7 Which world each block ran in

world	sim_variant	demo root	seeds / cells	pointer
old	base	baselines / matched_v 2 (N=56)	DP 10- 14 (dH/d DP/d R2D); RLPD 10- 13 (spar se+d ense, + fails arms spars e); all r2dr eam er cells (80- 93, 99- 108, 200- 203, 300- 303); N5/N 15/N 13/N	analy sis/ resul ts_ta ble.p y:10- 20; RESUL TS_TA BLE_2 026- 08- 25.md :3; UPDAT E:14; CRITI QUE_d ecisi ons:3 89- 391 (C3)

16

DP
20-
29;
RLPD
20-
29
(spar
se+d
ense)
; N7
densi
ty
20-
22;
N14
split
halve
s 40-
42;
r2d
pilot
60/61
(0/2
at TL
1200
)

```
SESSI
ON_LO
G:129
,141,
158-
160;
M/mat
ched_
w3/MA
TCHED
_SETS
.json
```

cor
rec
ted

```
gc_kp4_r
iser3_sh
elf6
(adopte
d by
user
08-24
~08:15
, shelf6
pendin
g the
real
shelf
measur
ement)
```

baselines
/
matched_w
3 (N=58;
dH +
dDP only,
no
corrected
-world
dR2D
teacher)

corrected + ts5

```
gc_kp4_r
iser3_sh
elf6_ts5
(commi
t
7c8195
d
"world
of
record
from
08-26")
```

matched_w
4_pilot/
dH only
(dH_w4
recording
08-26
07:50)

```
DP
pilot
dp_pil
otw4
only;
no
result
block;
matche
d_w4
proper
never
built;
no
kept
count
```

```
SESSI
ON_LO
G:153
-156;
resul
ts_ta
ble.p
y:19-
20;
POSTM
AINT_
COMMA
NDS.m
d:118
-126
```

/read
out
for
dH_w
4
recor
ded
in
any
note
**[CHE
CK
10]**

3. The MDP and the contract-v1 recorder (baselines/record_demos.py)

3.1 Environment

FullTaskEnv(backend='cpu', max_steps=1200, scope='pick', action_mode='delta_joint', delta_cap=0.025, delta_leash_mult=5.0, action_repeat=4, delta_ref='target', camera_rig=True, pick_shaping=False) (record_demos.py:94-103,219-231; baselines/rl/full_env.py:265-270,310). Asserted at construction: scope/mode/ref, repeat and cap, leash == 0.125, no shaping/hold reward, inner env never truncates (genv.max_steps = 1e9, the #26 bug guard) (record_demos.py:226-230; full_env.py:351). sim_variant applied pre/post build and stamped (:221,231,781).

3.2 Observation (17-dim state + images)

state = [q[0:6], gripper motor 0..1, grip effort, can xyz, can quat wxyz, goal xy] (genesis_can_env.py:298-313; pick_env.py:36). Gripper motor = $1 - (0 - (-0.09)) / (0.96 - (-0.09))$; grip effort = |control force| on the two bottom finger drivers ("sim analog of motor current"; its scale moves with the world variant: carry median 18.4 base → 26-28 ts5, gripper_lab:772-776). lerobot split: observation.state = states[:8] (proprio), observation.environment_state = states[8:] (can pose + goal) (convert_to_lerobot.py:81-89). Images (64,64,6) u8 top++wrist (§2.2); WM learners use images only (cnn_keys: image, mlp_keys: '\$^'), DP and RLPD use state only. **[CHECK 11]** genesis_can_env.py:10 docstring still says float32[16].

3.3 Action space and integrator

Box(-1,1,(7,)) (full_env.py:392). Per sim step (full_env.py:560-578):

```
d = clip(a[:6], -1, 1) * delta_cap # 0.025 rad / sim step
```

```

sp = clip(target + d, ARM_LO, ARM_HI) # delta_ref='target': integrate onto the
running target

target = q_meas + clip(sp - q_meas, -leash, +leash) # leash 0.125 rad around the measured
joints

grip_phys = (clip(a[6], -1, 1) + 1) / 2 # absolute 0..1

```

action_repeat=4: the same a is applied on 4 consecutive sim steps (advance $\leq 4 \cdot a \cdot \text{cap}$), rewards summed, loop breaks on terminated/truncated (full_env.py:531-542). Target re-seeds from measured q on every reset (:402-412). Calibration: cap 0.025 = the demos' p99 per-frame commanded delta ($44^\circ/\text{s}$ at saturation); leash $5 \times \text{cap} = 0.125 \approx$ the demos' PD-lead p99 (0.126) (full_env.py:300-302; paper/METHODOLOGY.md:233-244). Decision clock = 4 sim steps = 7.5 Hz. DP emits absolute window-end targets and is executed through the same integrator at eval ("hold-4", §6.2).

3.4 Horizon and termination

- max_steps = 1200 sim steps = 300 decisions; truncated = (not terminated) and $t \geq \text{max_steps}$ (full_env.py:689); the tape ends exactly at the terminal or the cap (record_demos.py:156,318-325).
- **Tip rule:** after each sim step, if grip_phys < GRIP_OPEN (0.3) and can tilt > TIP_DEG (60°) $\rightarrow$ terminated, info['tipped'], reward += TIP_PENALTY. **TIP_PENALTY = 0.0 for pick/full scope** (PLACE_TIP_PENALTY = -0.25 place scope only). PREREG §2 says $r=0$. The "-0.5" in the task brief/CLAUDE.md history is the July Cartesian env; not current **[CHECK 12]** (full_env.py:214-218,673-688,705). Census basis: past 60° the can never recovers (31/32); grip-open guard needed because demos carry the can pitched $>60^\circ$ in hand (CLAUDE.md; METHODOLOGY.md:120-124).
- The **eval** harness uses GenesisCanEnv (no tip termination, runs the full horizon) — §6.2.

3.5 Pick predicate ("hardened lift")

All of can_z > pick_z AND grip·100 > GP_CLOSE (30) AND $\|eef - can\| < \text{PICK_EEF_DIST}$ (0.20 m) sustained PICK_SUSTAIN = 10 consecutive sim steps; eef = wrist link position; pick_z = table_top + can_h/2 + 0.05 = 0.05 + 0.0505 + 0.05 = 0.1505 (genesis_can_env.py:48-65,255-265; replay_harness.py:184; census header "pick_z 0.1505"). Pick scope terminates on the sim step the grant lands (full_env.py:627-633); no settle. Hardening history and the frozen-policy fairness certification: paper/METHODOLOGY.md:109-118.

3.6 Rewards

- Sparse: STAGE_REWARD = {picked 1.0, placed 1.0, contact 2.0, nested 4.0}, paid once per first grant; pick scope $\rightarrow +1$ and terminate (full_env.py:57,593-

606,627-633). Tapes carry sparse env rewards only ("shaping is NOT baked in", `record_demos.py:22`).

- **Fast-pick 2.0 quirk:** if the pick grant and the placed grant land in the same repeat-4 window (`GenesisCanEnv.step` sets `_picked` then `_placed` in the same sim step when the can is already in the shelf footprint/z-band), the terminal row carries reward 2.0. Seen on r2d-teacher tapes (~68-step picks); `to_dreamer_native.py` accepts `{0,1,2}` and refuses positive reward before the terminal row; matched_v2 r2d/dR2D total_reward 64 for 56 tapes (8 tapes pay +2) (`baselines/rl/to_dreamer_native.py:65-72`; `genesis_can_env.py:265-267`; `SESSION_LOG:114`). Mechanism reading is from code, not from a comment **[CHECK 13]**.
- Dense (potential-based): $r' = r + \gamma \cdot \phi(s') - \phi(s)$, $\phi = -\text{PICK_SHAPING_SCALE} \cdot \|\text{eef} - \text{can}\|$ with `SCALE = 2.0`, $\phi(\text{terminal}) = 0$ (`pick_shaping_terminal_zero`), γ = the learner's discount (RLPD 0.998 via `pick_shaping_gamma=args.gamma`, asserted equal; r2d $1 - 1/\text{horizon} = 0.997$ after the patch; dv3 0.997); demo transitions relabelled offline from the recorded `eef_pos` with the same potential (`full_env.py:95-102,261-262,333,543-557`; `train_rlpd.py:99,245,255,308-312`; `cluster/patches/r2dreamer_final_rr.patch` hunks 1-2; PREREG §2).

3.7 terminal_from_tape and the terminal guard

`full_env.terminal_from_tape(tape, ...)` → `{t_term, kind ∈ pick|tip|other, reward, layout}`: contract-v1 tapes → reads terminated/picked/tipped (`full_env.py:152-168`); legacy stride-1 tapes → recomputes the pick proxy + tip rule, warns that the hardened eef distance is unavailable (`:173-211`). Every legacy encoder/converter has `--demo-terminal-guard` default **on** (tip → done, `r=0`, later frames dropped; pick → done; cap → bootstrap) (`train_rlpd.py:145-153`; `train_sacfd_full.py:622`; `to_dreamer_demos.py:70`; PREREG §4.3). This guard is the fix for the 08-19 dDP_RLPD 0/6 collapse (fail tapes without terminal flags → critic bootstrapped through states the env would have terminated; N1).

3.8 Tape contract v1 — keys and manifest

One npz per episode, filename = rollout index ≥ 100000 (`rollout_base + 1000 · shard_idx`, never a human uid; `record_demos.py:774-777,804`). Keys (`:16-33,329-341`): `states (n,17)`, `final_state (17,)`, `actions_delta (n,7)` (the executed normalized decision — the learners' action space), `actions (n,7)` (absolute window-end command: 6 joint targets rad + grip 0..1), `rewards`, `terminated`, `truncated`, `picked`, `placed`, `contact`, `nested`, `tipped (n,)`, `eef_pos (n+1,3)` (tool tip), `images (n+1,64,64,6)`, `sim_states (m,17)`, `sim_actions (m,7)` (per-sim-step sub-tape, $n \leq m \leq 4n$), `end_reason`, `n`. Scalars (`:115-120`): `uid`, `ic_uid`, `label ('success' iff picked on the last row)`, `stage`, `teacher`, `teacher_ckpt`, `act_mode`, `sim_variant`, `action_repeat=4`, `delta_cap`, `delta_leash`, `delta_ref='target'`, `pick_z`, `n`, `git_sha`, `env_class='FullTaskEnv'`, `recorder='record_demos.py v1'`, `contract='v1'`, `end_reason`, `scope`, `max_sim_steps`, `verify`. Validator: exactly one of `terminated`/`truncated` on the last row and never earlier (`:125-200`). Manifest (`:830-846`, merged `:599-635`): full argparse config incl. `tol`, `max_dwell`, `arrival`, `dilation_cap`, `settle`, `attempts`, `verify`, `seed`, `shard`, `git_sha`, `teacher_ckpt`; `n_rollouts`, `n_kept`, `n_fail`, `rejected_by_verify`, `teacher_success_rate_first_attempt`, `yield_frac`; per-rollout records (`ic_uid`, `ic_can_xy`,

outcome, decisions, sim_steps, seconds, kept, end_reason, attempt, verify, dilation...); content_sha256 (sha over sorted name bytes + file bytes), built. Fails go to <outdir>_fails/ (same schema, label='fail'); the 5xxxxx stems are assigned by the set builder (make_matched_sets.py:151-161: 500000 + stem % 100000), 6xxxxx by make_succ_control_set.py.

3.9 Adapters and guards

adapter	behaviour	pointer
human	§2.4; attempts forced to 1; verify off (the keep predicate is the pick itself)	record_demos.py: 395- 470,694- 695,742
dp	lerobot DP queried once per decision on the current obs; absolute target $q^* \rightarrow$ a_arm = clip(($q^* - \text{target_now}$)/(4·cap)); grip = chunk value ·2−1; --mode sample (default) / mode (reseeds per episode); policy on GPU if visible, sim CPU (A4); -- attempts 3 --verify	:473- 504,654, 663; manifests
r2d	champion loader, native delta actions, asserts cfg action_mode/repeat/cap/leash/size; -- mode sample --attempts 3 --verify; -- torch-threads 2	:507-582
random	uniform $[-1,1]^7$ per decision (rng seed+7919); warns if kept > n/30	:388- 392,849- 850
-- verify	open-loop replay of actions_delta from a fresh reset of the same IC must re-earn the pick; failures routed to fails with outcome='verify_failed'; pick scope only	:362- 371,685- 687,798- 803
--ic- from- tape	§2.3	:662,731- 757
--sim- variant	default base; merge refuses mixed variants	:669,619- 622

4. Demonstration sources and sets

4.1 Recordings (all by `record_demos.py v1`, repeat 4, cap 0.025, leash 0.125, 1200 sim steps, `src = baselines/episodes_pick_phase_dppruned, --ic-from-tape`, 66 ICs cycled)

recordin g	w o r l d	teacher / adapter	rollout s → kept	first- attemp t rate
dH (meas)	b a s e	human, --arrival meas, git 91cd7b1/ bde726d	66 → 51	—
dH_either	b a s e	human, --arrival either, git 33d5bc9	66 → 56	0.848
dDP	b a s e	dp, ckpt baselines /outputs/ dp_pilot/ dH_DP_s0/ checkpoin ts/020000 /pretrain ed_model (DP-r4 pilot s0, sel 13/15;	80 → 64	0.818

		PREREG		
		§3.1		
		"median		
		in-dist		
		seed" →		
		with 2		
		pilot		
		seeds		
		the		
		lower), --		
		mode		
		sample --		
		attempts		
		3 --		
		verify,		
		git		
		1dcd268		
			16 → 14	
			files: 8	
			env_trun	
			cated at	
			300	
			decision	
			s (uids	
			258 266	
			267 298	
			299 304	
dDP_fails	b	same	326	0.0
	a	teacher	327) +	
	s		6 lying-	
	e		can	
			tapes of	
			1	
			decision	
			(234×3,	
			318×3);	
			2 verify-	
			rejected	
dR2Dprov	b	r2d, ckpt	73 → 64	not
	a	C/r2dream	(8	stampe
	s	er/runs/p	tipped,	d
	e	ick_v5d4c	1 verify-	
		_delta_sh	failed	
		aped_dH_s	uid 333)	
		51/BEST_s		
		elected.p		
		t		

		(PREREG A2 fallback, not CHAMPIO N_15768 20.pt), -- mode sample -- attempts 3 -- verify, git 3167d12/ f6d937f		
_smoke_neg ctl	b a s e	random, n=3	3 → 0 (2 truncate d, 1 tipped)	0.0
dH_w2	g c – k p 4 – r i s e r 3 – s h e l f	human, either, git e306ec9	66 → 58 (drops: tipped 234, 318; env_tru ncated 245 246 286 293 295 300); dilation p50 0.99, max 1.07	0.879

	6			
	g			
	c			
	-	dp, ckpt		
	k	baselines		
	p	/outputs/		
	4	dp_pilotw		
	-	2/dH_DP_s		
	r	0/checkpo		
	i	ints/1000	76 → 64	
	s	00/pretra	(7	
dDP_w2	e	ined_mode	tipped,	0.879
	r	1 (pilot	5	
	3	hold	truncate	
	-	14/15,	d)	
	s	rnd		
	h	18/30),		
	e	attempts		
	l	3 --		
	f	verify,		
	6	git		
		2061f08		
		does		
		not		
		exist (no		
		corrected		
		-world		
		r2d		
		teacher;		
		the		
corrected		matched_w		
-world	—	3/dR2D		
dR2D		row was		
		a byte-		
		identical		
		copy of		
		dDP and		
		was		
		purged,		
		BL-7)		
dHunpruned	b	human	recorde	
{,_either}	a	on	d; no	
	s	episodes_	result	
	e	pickphas	appear	
		e_all	s in	
		(control		

for §4.5
pruning)

any
results
file
[CHECK
14]

dH drop reasons vs PREREG A1 wording ("2 lying-can, 1 knock-over, 5 over-horizon, 7 over-cap" for the 15 meas drops): recorder outcomes are 3 tipped (234, 318 lying-can ICs; 246 knock-over), 10 env_truncated (1200-step horizon), 2 adapter_exhausted (dilation cap 3.0: 257, 298) — the 5/7 split does not match the 10/2 outcome codes [CHECK 15] (recorder_manifest_dH.json records[]; PREREG:241-243). Lying-can ICs 234 and 318 are the two human demos whose recorded can starts on its side; they are unachievable in the MDP and also make uid 234 the sel ceiling (§6.1).

Teacher circularity (disclosure): both model teachers descend from dH (dDP = DP trained on the 51-tape meas dH; dR2D = r2dreamer dense trained on the 08-19 human set) (PREREG §3.1, §10).

4.2 Matched sets (baselines/make_matched_sets.py)

Rule: $N = \min(\text{success counts}, \text{--cap-n } 66)$; stratified round-robin over ic_uid with common capacity $c(u) = \min \text{ over sources}$ so all sets cover the same uids with the same multiplicity (ic_multiset_identical: true); rng = default_rng(seed=0); tapes hard-linked; refuses non-v1 tapes, label/picked mismatch, unterminated tapes, mixed sim_variant (:64-78,103-142,167,183,207,211). Frames are not matched (intrinsic; reported). content_sha256 over sorted name + bytes (:94-100); every sbatch asserts the sha of the set it consumes. Fails arms: $F \leq \lfloor 0.30 \cdot N / 0.70 \rfloor$; lying-can fail tapes ($n \leq 1$ and tipped) excluded since commit ca05472 (W5) — hence matched_v1 fails arms carry 14 fails and matched_v2 carry 8 (:222-231). Per-learner derivations are built separately: lerobot via convert_to_lerobot.py <set> <set>/lerobot 8 4 none image (fps = $30/4 = 7.5$, gated against meta/info.json), dreamer-native via to_dreamer_native.py (§4.7); sbatch_dp.sh:14,221 wrongly says make_matched_sets builds them [CHECK 16].

root	built / git	N	dH sha	dDP sha	dR2D sha
matched_v1	08-23 13:53 / 0e1fda19	5 1	58f d8 9b c	9be 552 19	fe9 3d bd d

matc hed_ v2 (blo ck of reco rd, old worl d)	08- 23 14:1 7 / ca05 4726	5 6	4d 4e a0 8f	299 825 ad	10c bc9 b2
--	--	----------------------	----------------------	------------------	------------------

matc hed_ w3 (cor rect ed worl d)	08- 24 11:4 6 / 2061 f082	5 8	07 ce 3f3 6	d3b f95f 6	—
--	--	----------------------	----------------------	------------------	---

matc hed_ w4	neve r built (only match ed_w4 _pilo	—
--------------------	--	---

t/dH)

Provenance chain for the WM fails arm (state once): RLPD manifest sha b222df06... = repeat.json.src_sha → native sha 3a6df556... → registry fingerprint c5ae239b... (AUDIT_results_2026-08-28.md §5).

4.3 Census (analysis/characterize_demo_sets.py; constants pick_z 0.1505, cap 0.025, leash 0.125, tip 60°/grip<0.3)

metric	v1 dH / dD P / dR 2D	v2 dH / dD P / dR 2D	v2 dDPfail s / dR2DDP fails	w3 dH / dD P	po in te r
tape s	51	56	64 / 64 (8 fails each)	58	ce ns us .m d pe r ro ot
tran sitio ns (dec isio ns)	64 91 / 62 94 / 86 1	70 02 / 69 09 / 95 4	9309 / 3354	69 27 / 72 85	
tape leng th p50 (dec isio ns)	11 3 / 11 3 / 17	11 5 / 11 4 / 17	122 / 17 (fail len 300)	11 0 / 12 0	

	12	12		
	7,2	5,2		
tape	85	84		11
leng	/	/		9,2
th	12	12	145,300	46
mea	3,2	3,2	/ 52,300	/
n /	33	33		12
max	/	/		6,2
	17,	17,		35
	24	24		

fail
share
of
buffer
(rows)

			0.258 /	
			0.716	
			(=25.8	
			% / 71.6	
			%;	
—	0		PAPER_N	0
			OTES'	
			74/28	
			was	
			wrong)	

unique
ICs /
grid
cells
(of
16)

50	55		57
/	/	56 / 13	/
13	13		13

reward
density

.00	.00		.00
79	80		.00
/ .0	/ .0		84
08	08	.0060 / .	/ .0
1 /	1 /	0167	08
.05	.05		0
92	87		

decision
s at

.05	.06		.05
5 /	9 /		4 /
.01	.01		.02

AU
DI
T_
re
su
lt
s:
96;
M/
ma
tc
he
d_
v2
/c
en
su
s.
md

the cap	0 / .04 8	0 / .04 7			4
hum an time dilat ion p50 / max	(bl an k)	1.0 3 / 1.6 5			0.9 9 / 1.0 7
tape s tipp ed at t0	0 (fai ls ar ms 6)	0	0		0
all tape s: cont ract v1, repe at sta mp 4, ter min al on last row only , ima ges pres ent	yes	yes	yes		yes

The "decisions per tape" summary used in prose (dR2D ~17, dDP ~123, dH ~115) is the v2 p50/mean mix. Fail tapes are 300-301 rows each (cap-truncated, bootstrapped), 12× the median dR2D success tape and 3× the WM training

time_limit; they are 70 % of the fails arm's demo rows (AUDIT_results §1).

4.4 Split halves (baselines/make_split_halves.py, N14)

Group tapes by `ic_uid`, shuffle uid list with `default_rng(seed=0)`, alternate whole ICs A/B; per-half manifest with `split_half`, `split_seed`, `sha (:28-56)`. Run on matched_w3 dH and dDP: "58 tapes over 58 ICs → A 29 / B 29" (SESSION_LOG:157). Halves' shas not captured locally **[CHECK 17]**.

4.5 Pruning

- Leading-idle pruning applied ONCE to the human stride-1 tapes before re-recording (`make_dp_pruned.py`: collapse idle runs with action delta < 1e-3 before `j_pick` - 150 frames; 29.6 % of frames) — this is `baselines/episodes_pick_phase_dp_pruned`, the src of every recording; the unpruned control is `dHunpruned` (`make_dp_pruned.py`:7-15,75-81; PAPER_PLAN.md:400-407; PREREG §3.1).
- Post-hoc density control sets (N7, `baselines/make_pruned_bc_set.py`:46-49): keep decision iff $\max|a_{\text{arm}}| \geq \text{eps}$ OR $|\Delta a_{\text{grip}}| > 5e-3$; terminal always kept; BC-only (sim sub-tape dropped). Measured at eps 1e-3 on matched_w3/dH: 6927 → 6116 decisions (**11.7 % removed**; 414 of the removed had grip closed). The 1e-2 fraction ("22 %") exists in no artifact — source or delete **[CHECK 18]** (SESSION_LOG:159; AUDIT_results:84,114).

4.6 Fails arms and controls

- `dDPfails` = 56 dDP successes + 8 dDP fails (episode share 0.125; row share 25.8 %); `dR2DDPfails` = 56 dR2D successes + the same 8 fails (row share 71.6 %). The 8 fail tapes are byte-identical across the RLPD and WM arms (AUDIT_results §0).
- N16 controls (`make_succ_control_set.py`, PREREG A10): `dR2DDPsucc` = dR2D + the 8 longest dDP success tapes (6xxxxx stems; 1521 added rows vs the fails' 2400 → `length_match_fraction` 0.634; N 64) (`M/matched_v2/dR2DDPsucc/manifest.json`); `dR2Ddup13` = dR2D at `demo_duplicate 13` (ring share $\approx 27\%$ = fails arm) via `sbatch_r2dreamer.sh`:199,236. Predictions registered in PAPER_NOTES.md N16 (:419-441).
- Exposure arithmetic of record (supersedes every "1-3 %" / "1/17th" figure): with `demo_reinject_every 150000` × `demo_duplicate 4` over a 3e6-step budget → 19 re-injections; steady-state ring share dR2D $\approx 9.8\%$, fails arm $\approx 26.7\%$ (fail rows 18.8 %); RLPD 50 % demo per batch, fail rows 35.8 % → exposure ratio $\approx 2\times$ (AUDIT_results §1).

UPDATE 08-29. Same-source fails arms only (user rule 08-28: never mix sources): `dHHfails` = 56 dH successes + 8 dH fail tapes; `dDPfails` = 56 dDP successes + 8 dDP fail tapes (`baselines/make_samesource_fails_arm.py`; fail share 0.30 by tape was requested but only 8 fail tapes per source survive the lying-can/success-label filters → share 0.125 by tape, fail rows 2145 / ~2300 = 23-25 % of rows; 7 truncated + 1 tip-terminated (dH), 8 truncated (dDP)). Row-matched same-source controls

dHsucc_dup / dDPsucc_dup = the 56 successes + the 8 LONGEST own successes duplicated (6xxxxx stems; baselines/make_succ_control_set.py --donor <same arm>; added rows 1755 / 1521). Reading rule (A17): a fails effect is claimed only if +fails differs from +succ_dup in the same direction on both sources. Census per source in paper/tape_census_2026-08-29.txt (baselines/diagnostics/tape_census.py): dH vs dDP identical on N, rows, terminal rows, saturation ($\leq 6\%$ on any joint), state ranges; dDP tapes carry verify=pass (open-loop replay at record time), dH verify=n/a. The mixed-source dR2DDPfails / dR2DDPsucc arms are mechanism cells only (N15, A12).

4.7 Dreamer-native conversion (baselines/rl/to_dreamer_native.py)

action[t] = actions_delta[t-1] (zeros at 0); reward[t] = rewards[t-1] × terminal_reward (zeros at 0); is_terminal[-1] = terminated[-1] (cap-truncated → False → bootstraps); discount = 1 - is_terminal; is_first/is_last; images T = n+1; refuses shaping, rewards $\notin \{0,1,2\}$, positive reward before the terminal, repeat-stamp mismatch, mixed cap; writes repeat.json (action_repeat, contract, delta_cap, scope, terminal_reward, src_sha, src_manifest_sha, census) (:49-83,111-131,146-156). r2dreamer dirs built with --terminal-reward 1 (prefill multiplies by reward_scale 100, asserted); dv3 dirs with --terminal-reward 100 — same effect, different file (SESSION_LOG:100,111; r2dreamer_final_rr.patch hunk 3).

5. Learners

5.1 DP — lerobot Diffusion Policy (state-only)

knob	value	pointer
Code	lerobot fork dirkmcpherson/lerobot branch genesis-fixes (lerobot 0.4.5 + image_writer mkdir fix; cluster checkout b63920e), torchcodec 0.3.*	cluster/ install_l erobot.sh :17- 26,49,60- 68
	observation.state (8) + observation.environment _state (9); no images (cameras none)	sbatch_dp .sh:82- 85,420; convert_t o_lerobot .py:81-90
Action	7 = 6 absolute joint targets (window end) + grip 0..1; fps 7.5	convert_t o_lerobot .py:42- 51,90

Train

```
lerobot-train --  
policy.type=diffusion  
--seed=$SEED --  
batch_size=64 --  
steps=100000 --  
save_freq=20000
```

```
sbatch_dp  
.sh:187,1  
98,203-  
204,448-  
454
```

Diffusion
Config
(lerobot
defaults,
nothing
override
n)

```
n_obs_steps 2; horizon  
16; n_action_steps 8;  
drop_n_last_frames 7;  
normalization  
MIN_MAX (state,  
action); UNet  
down_dims  
(512,1024,2048),  
kernel 5, n_groups 8,  
diffusion_step_embed  
128, FiLM; DDPM 100  
train timesteps,  
squaredcos_cap_v2,  $\beta$  1e-  
4→0.02,  $\epsilon$ -prediction,  
clip_sample 1.0;  
num_inference_steps  
= 100; Adam lr 1e-4,  
betas (0.95, 0.999),  
wd 1e-6; cosine  
schedule, warmup  
500; vision backbone  
not instantiated
```

```
cluster  
fork  
src/lerob  
ot/polici  
es/diffus  
ion/confi  
guration_  
diffusion  
.py:104-  
163;  
modeling_  
diffusion  
.py:204-  
207
```

Checkpoi
nts

```
K=5 at 20k...100k,  
each with  
dp_sidecar.json  
(action_repeat 4,  
demo_sha,  
sim_variant, git, node,  
config)
```

```
sbatch_dp  
.sh:459-  
482
```

Execution

```
hold-4 through the env  
integrator (§3.3);  
chunk of 8 decisions  
between replans;  
sampled actions,  
diffusion noise seeded  
per episode (--seed k);  
policy on GPU if visible
```

```
wandb_eva  
l.py:105-  
109,271-  
326;  
dp_runner  
.py:39-  
50;  
eval_swee  
p.sh:118-  
128
```

Selection	cluster/ dp_select_confirm.sh: score 5 ckpts on sel, best (ties → later, -ge) confirmed on hold+rnd, final also on hold+rnd; DP-HEADLINE line + sweep/HEADLINE.txt	dp_select_confirm. sh:23-50; PREREG A7
Seeds	old world (matched_v2): dH/dDP/dR2D × 10-14 (n=5); corrected (matched_w3): dH/dDP × 20-29 (n=10); N7 density arms 20-22; N14 halves 40-42; pilots s0/s1	M/ RUN_REGIS TRY.jsonl ; results_t able.py:1 0-20
Not run	the PREREG §1 300k "epoch-matched view" (no registry row, no sbatch path) [CHECK 19]	PREREG:32
Resources	`-p gpu,preempt -- requeue --gres=gpu:1 --constraint="l40s\	a100" -n 8 --mem 48g -- time 14h`

sbatch_154-167

5.2 RLPD — SB3 SAC subclass (baselines/rl/train_rlpd.py, rlpd_sac.py, cluster/sbatch_rlpd.sh)

knob	value	pointe
------	-------	--------

		r
Base	stable-baselines3 2.8.0 (asserted)	rlpd _sac .py: 38- 40; sbat ch_r lpd. sh:2 40
Critic ensemble	E = 10 LayerNorm critics (vectorised EnsembleLinear), LN after each hidden Linear, shared affine (--per-member- ln off)	trai n_rl pd.p y:58 ; rlpd _sac .py: 91- 126, 372
Target subset	Z = 2 random target critics, min → Bellman target, redrawn every grad step	trai n_rl pd.p y:60 ; rlpd _sac .py: 258- 260
Networks	π (256,256), Q (256,256)	rlpd _sac .py: 345, 371

UTD	10 critic grad steps per decision; actor + α once per decision	<pre> train_rlpd.py:55 ; train_rlpd_sac.py: 240- 241, 282- 301, 363- 364 </pre>
Batch	256 = 128 online + 128 demo (symmetric sampling from an immutable demo buffer)	<pre> train_rlpd.py:61 ; train_rlpd_sac.py: 232, 243- 249, 360 </pre>
γ	0.998	<pre> train_rlpd.py:99 </pre>
Entropy	ent_coef auto, target $-\dim/2 = -3.5$, entropy backup off	<pre> train_rlpd.py:63 ,103 ,107 ; train_rlpd_sac.py: 269- 271, 350- 352 </pre>
lr / τ / buffer / learning_ starts	3e-4 / 0.005 / 300 000 online / 1 000	<pre> train_rlpd_sac.py: 357- 361 </pre>

Budget	100 000 decisions = 400 000 sim steps (BUDGET_UNIT=decision s); train horizon 1200 sim steps; eval horizon 1200	sbat ch_r lpd. sh:3 3,80 - 83,8 9
Env	FullTaskEnv(scope='pick', action_mode='delta_joint', action_repeat=4, delta_ref='target', pick_shaping=..., pick_shaping_gamma=0.998, pick_shaping_terminal_zero=True), cap 0.025, leash 0.125	tra n_rl pd.p y:23 8- 246
Demo format	--demo-format native: (states[t], actions_delta[t], rewards[t] (+shaping from eef_pos when dense), next = states[t+1] or final_state, done = terminated[t]); truncation bootstraps; sim_variant/repeat/ cap/leash/delta_ref stamps checked; -- demo-terminal-guard on for legacy tapes only	tra n_rl pd.p y:14 5- 160, 307- 312; sbat ch_r lpd. sh:1 91
Dense	--pick-shaping on -- demo-shaping on -- pick-shaping- terminal-zero on (\$3.6)	sbat ch_r lpd. sh:1 86- 190
Demo RNG	torch.Generator().manual_seed(seed) (post-2fbed2a fix)	rlpd _sac .py: 179- 182

Check points	K=5 via ArchiveCheckpointCallback at 0.2...1.0 of steps → ckpt_020... ckpt_100/rlpd_ckpt.zip ip + .action_mode.json sidecar; rlpd_final.zip	train_rlpd.py:17 1-175, 440-471, 479-480		
Eval action	deterministic (predict(deterministic=True)), policy on CPU	wandb_eval.py:23 6,26 3; eval_sweep.sh:12 6		
Select ion	same as DP (5 × sel → best, ties → later → hold+rnd; SWEEP-HEADLINE)	sbatch_rlpd.sh:2 48-306		
Seeds	old world (matched_v2): dH/dDP/dR2D × sparse+dense × 10-13 (n=4); dDPfails, dR2DDPfails sparse 10-13; corrected (matched_w3): dH/dDP × sparse+dense × 20-29 (n=10). (Task brief's "10-14" applies to DP, not RLPD [CHECK 20])	registry		
Resou rces	`-p gpu,preempt -- requeue -- gres=gpu:1 -- constraint="l40s\	a10 0\	l 4 0 \	h 2 0 0 "

-
n

8

-
-
m
e
m

4
8
g

-
-
t
i
m
e

1
2
h
、

		PRE
		RE
		G
Positi	none passing	\$10
ve	(PREREG \$10	
contro	disclosure); recipe	;
l	tuned under old-	PAPE
	world dynamics	R_NO
	(N6/N10)	TES.
		md
		N6

UPDATE 08-29 — recipe restart (PREREG A17). All $\gamma = 0.998$ runs (seeds 10-19, 24-29; waves `final`, `w2final`) are superseded, diagnostic-only (E9): 69/74 diverged (critic loss 10^2 - 10^5 , Q-watchdog trips, hold $\leq 3/15$). Fix factorial (seeds 30-32, old world, sparse): γ 0.99/UTD 10 (**g99**) — dH 3/3 stable (max critic loss 0.016-0.027, 0 trips, hold 14/14/14, rnd 20/21/19); dDP hold 10/13/14, rnd 14/12/20 with 2/3 seeds diverging (max critic loss 22, 84). γ 0.99/UTD 5 same picture; γ 0.998/UTD 5 diverges $\rightarrow \gamma$ is the cause. **Matrix recipe = γ 0.99, UTD 10, E 10, Z 2, 50/50 demo batches, LayerNorm critics, `ent_coef` auto, target entropy -3.5** (Ball et al. 2023 values except UTD 10 vs 20). Seeds (fresh ids, A9):

dH/dDP sparse 30-37; dHHfails/dDPfails sparse 30-37; dH/dDP dense 30-35; dHsucc_dup/dDPsucc_dup sparse 30-33. Launcher knobs `GAMMA`, `UTD` (`cluster/sbatch_rlpd.sh`, registry keys). Dense arms relabel demo rows with the run's γ at load (`train_rlpd.py:311`) — γ 0.998 and 0.99 dense runs never share a table. Q-watchdog is warn-only (`rlpd_sac.py:318-324`) and its 2.0 threshold is invalid on dense arms (potential shaping lifts Q); health on dense arms = max critic loss + final $\approx$ selected. Statistic (A16, pre-registered before any s33-37 readout): divergence rate (Fisher) + LAST-checkpoint rnd-30 success (Welch + exact permutation, `analysis/stats.py`) — result 08-29 20:00: LAST rnd dH 0.700 vs dDP 0.494 (+0.206, CI [+0.105, +0.306], p 0.001), divergence 1/8 vs 6/8 (Fisher 0.041), see `RESULTS_for_writing §2`; hold is reported, not the statistic (dH at ceiling 14/15; hold $\subset$ training ICs, A8). Time limit 18 h (one 12 h job lost its final-ckpt eval on a slow node). Env fix timing: jobs started before 08-28 13:25 carried the +2 double pick grant (`full_env.py`, `n_double_grant` normalised in the native loaders) — g99 runs are all post-fix.

5.3 r2dreamer (world model, pixels)

What it is: R2-Dreamer (Morihiro et al., ICLR 2026) — a decoder-free, augmentation-free DreamerV3-family world model with a Barlow-Twins-style redundancy-reduction representation loss, shipped with its own PyTorch DreamerV3 reproduction (`model.rep_loss` $\in$ {`r2dreamer`, `dreamer`, `inforce`, `dreamerpro`}); vs DreamerV3: no decoder, BlockGRU (8 blocks), barlow loss 0.05 (`lambda` 5e-4), and a **second, replay-based critic loss** (`repval`, weight 0.3) that backs up a λ -return along the recorded trajectory (demo/fail tapes included) bootstrapped with the imagined return (`C/r2dreamer/README.md`; `_base_.yaml:23,36,45,51,76`; `dreamer.py:539-563`, `disc :549`, `clamp :552-553`; `AUDIT_results §1`). Port: cluster tree at commit `c25eb3b` + working-tree edits captured in `cluster/patches/r2dreamer_final_rr.patch` and `cluster/r2dreamer_port.tar.gz` (`AUDIT_sources_2026-08-23.md:248-287`).

Patch (`cluster/patches/r2dreamer_final_rr.patch`): (1) `envs/genesis.py`: shaping γ from a ctor kwarg (`pick_shaping_gamma`, was hard-coded 0.999) with $\phi(\text{terminal})=0$, plus `R2D_SIM_VARIANT` hook; (2) `envs/_init_.py`: passes `1 - 1/horizon = 0.997`; (3) `demo_prefill.py`: native stride-4 ingestion (`repeat.json` stamp must equal `action_repeat`, `demo_downsample=1`, terminal reward `1.0 × reward_scale`); (4) `eval_genesis.py`: `--ic-file/--ic-set/--ic-index`, labelled outputs.

kno b	value	point er
Con fig	<code>configs/ env/ genesis_pic k_v5d4c_del ta_shaped.y aml (dense; sparse =</code>	<code>cluster yaml; sbatch_ r2dream er.sh:1 90</code>

	same without _shaped)	
Budget	3e6 sim steps (incl. prefill) = 750k decisions; 6 CPU worlds	yaml : 33-34
action_report / action space	4; delta_joint cap 0.025, leash mult 5; actor bounded_normal_clipped (min_std 0.1, max_std 1.0)	yaml : 36,46-51; _base_.yaml:17 2-174; METHOD0 LOGY.md \$7.3
time_limit	400 sim steps (100 decisions)) for every result cell (TIME_LIMIT=400); the PREREG \$8 pilot at 1200 failed (0/2 seeds at 2e6) → PREREG's stated on-fail action "keep 400" — amendment not formally	yaml : 40; sbatch_r2dream er.sh:2 40; CRITIQU E_decisions Q6

	logged [CHECK 21]	
Replay	buffer.max_ size 450000 rows, FIFO (LazyTensor Storage); demo_duplic ate 4; demo_reinje ct_every 150000 online steps (→ 19 re- injections/ run; ring share ≈10 % / ≈27 % fails arm); buffer not persisted across requeue	sbatch_ r2dream er.sh:3 07; yaml : 59-60; buffer. py:15; AUDIT_r esults §1
reward_ scale	100 (env and demo)	yaml : 61
train_ ratio	512 ; batch 16 × 64; pretrain 1000	yaml : 39; configs .yaml:1 7-18,30
horizon → discount	333 → 0.997 ; imag_horiz on 15; λ 0.95; return_cla mp 100	yaml : 65,82; _base_. yaml:17 ,19; dreamer .py:494 ,502- 503,549
acton	0 (demos)	yaml :

r_bc _lam bda	consumed as data only, by design)	52
act_ entr opy	3e-5	yaml : 66
Loss es	barlow 0.05, rew 1, con 1, dyn 1, rep 0.1, policy 1, value 1, repval 0.3; kl_free 1.0; unimix 0.01	_base_. yaml:16 ,35- 48,79
Mod el	size12M: deter 2048, hidden 256, discrete 16, stoch 32, cnn depth 16, units 256, SiLU, norm; symexp- twohot 255-bin reward & critic heads; CNN encoder only (cnn_keys: image)	configs / model/ size12M .yaml; _base_. yaml:68 - 81,134- 146,185 -197; yaml : 68-73
Opti	lr 4e-5,	_base_.

m	AGC 0.3, warmup 1000, Adam (0.9, 0.999); slow target update every step, fraction 0.02	yaml:25 -33	
Dem o sets	baselines/ matched_v2/ r2d/<ARM> (dreamer- native, terminal reward 1), stamp gate	sbatch_ r2dream er.sh:2 04-259	
Sele ctio n	every latest.pt archived and scored in- job on sel (15 episodes in one process, --mode sample -- seed 0 -- device cpu) → ckpt_scores .tsv; `BEST = sort -rn	head - 1 → BEST_ select ed.pt (byte copy, confirm ed); pruning keeps best-2 + newest + K=5 fractio n ckpts. Tie rule is sort`- order, not "later" [CHEC	sbatch _r2dre amer.s h:444- 548

Prot
ocol
read
out

cluster/
r2d_rescore
.sh:
BEST_select
ed.pt on
hold and
rnd, one
fresh
process
per
episode,
CPU, PAR=4,
--max-steps
1200,
asserted
denominat
ors, non-
zero exit
on missing
→ RESCORE-
RESULT
(replaces
an inline
loop that
printed
DONE over
540
crashed
processes)

r2d_res
core.sh
:1-53;
commi
t
bf09060

See
ds
(all
old
worl
d,
mat
che
d_v2
,
dens
e, TL
400)

dR2D 80-
93;
dR2DDPfai
ls 80-93
(s83 lost,
A9); dH
99-108
(s102 →
s108); dDP
100-107;
controls
dR2Ddup1
3 200-
203,
dR2DDPsu
cc 300-

registr
y;
SESSION
_LOG:16
7,173,1
79;
N16

	303; earlier 08- 19 cells dH 50-53 (dense), dR2D 40- 43 (sparse); corrected- world pilot 60/61 (TL 1200)			
Not run	PREREG §1 sparse arm and 6 seeds per source at TL 1200 (dense only, TL 400 ran) [CHECK 21]	PREREG: 35		
Res ources	`-p gpu,preem pt -- requeue -- gres=gpu: 1 -- constraint ="l40s\	a100\	l40\	h20 0" -n 8 -- me m 48g -- time 2d (rel aunc hes with -- time 24:0 0:00 `); sepa rate py3. 11 ven

UPDATE 08-29 — what is standard. The replay-buffer critic loss (`loss_scales.repval 0.3`, `dreamer.py:539-563`) matches the official DreamerV3 v2 recipe (`beta_repval 0.3`, `danijar/dreamerv3`; `paper/WM_CANDIDATES_2026-08-29.md`); only the imagined-return clamp (`return_clamp 100`, `dreamer.py:502-503`, `552-553`) is a deviation (E8 corrected). **Standard arm = `env.return_clamp=0`, `repval kept`** (runs `pick_v5d4c_delta_shaped_NOCLAMP_{dH,dDP}_s12{2,3}`, packed) — result 08-29 22:00: value runaway (train/val 115–825 vs max 100 at ~0.7M); the clamp is load-bearing. The earlier "STD" attempt (clamp off + repval off, `s120/121`) died at launch on a non-existent hydra key and is void. LAST-checkpoint hold re-scores (21 runs, `baselines/outputs/n12_rescore/_LAST_hold_`): 0/15 or $\geq 10/15$ in 16/21 — final policies are bistable; BEST-of-K selection on `sel` (training ICs) does the work and must be disclosed as such. `dup13` pilot (`s202/203`, `sel-only`): 0.93 / 0.00. Corrected-world gate pairs `dH/dDP s80-83` (`DEMOSET=w3`, `gc_kp4_riser3_shelf6`, dense, 3M): **ignites** — `sel` max per seed `dH` 0.40/0.93/0.93/0.93, `dDP` 0.00/0.87/0.13/0.93, bistable across adjacent checkpoints as in the old world; protocol re-scores (BEST hold/rnd, LAST hold; tags `_w3`): `dH` BEST hold 10/15/11/10, rnd 20/25/22/16; `dDP` 1/2/2/15, 1/2/3/28 (`n 4 v 4`, direction human > DP, perm p 0.143); seeds 84–87 per source added 08-29 (4-seed packs, `-n 32`). Under A19 the WM arm reports in one named world per table; the `w3` run dirs carry no world in their name (seeds 80–83 of `dH/dDP` = corrected world only). Packing: `cluster/sbatch_r2dreamer_pack.sh` (2 runs/GPU, 8 threads each, 16 CPUs, 96 GB; >2× per-GPU throughput, same launcher/config per run; 4/GPU with `-n 32` on 64-core nodes from 08-29). Touchgoal probe (no demos, `DEMO_DIR=null` now accepted by the launcher) `s0/1` packed.

5.4 dv3 — DreamerV3 (`dreamerv3-torch-genesis`) — documented negative

Patch `cluster/patches/dreamerv3-torch-genesis_final_rr.patch`: converter refuses `contract-v1` dirs / `--terminal-guard on`; `genesis_eval.py` gains `--ic-file/--ic-set/--ic-index/--device` and records `act_selection='deterministic'`; overlay `genesis_final_rr`: `time_limit 1200`, `steps 1e6`; new `sbatch_genesis_final_rr.sh` (`K=5` archive, fresh-process eval `sel` → hold+rnd, DV3-RESULT). Config: `action_repeat 4`; `time_limit 1200`; `steps 1e6` sim (250k decisions = 62.5k updates); `train_ratio 256`; batch 16 × 64; prefill 2500 / pretrain 100; `reward_scale 100`; `discount 0.997`, λ 0.95, `imag_horizon 15`; RSSM deter 512, hidden 512, `stoch 32×32`, GRU blocks 8, units 512; `model_lr 1e-4`; dense variant `genesis_pick_shaping: True` γ 0.997 (`C/dreamerv3-torch/configs.yaml:40-58,94,101-104,384-492`). **Status: the final-RR dv3 block never ran** (registry: 0 rows of `sbatch_genesis_final_rr.sh`; one 2e4 smoke). The dv3 negative (N4: transient takeoffs at 140–320k, no sustained ignition) rests on the 08-19 `sbatch_genesis_multi.sh` 300k runs (`dH s0-2`, `dDP s0-1`, `dR2D s0-1`), whose harness is outside this repo **[CHECK 23]** (`PAPER_NOTES.md:87-92`; `CRITIQUE_launch_plan_2026-08-25.md:415`; `PREREG:37`). Earlier history: unnormalised demo actions poisoned runs v6–v13 (`METHODOLOGY.md §6.3`).

UPDATE 08-29 — no longer "documented negative"; mechanism + standard arm. Diagnosis (`paper/DV3_DIAGNOSIS_2026-08-28.md`): on every unclamped run `value_mean` runs past the attainable return (dense ×100: 263–390 by 300k on

3/3 5M baselines; touchgoal-from-scratch s0 learned to 0.82–0.96 success at 230–440k then collapsed as value crossed 100) — critic runaway via bootstrapped reward-head leak; the torch port (NM512, pre-v2) lacks the official replay critic loss / EMA regulariser. Arms (dH dense, seeds 20–22 at 1M; launcher EXTRA_CONFIGS/TAGSUF): **genesis_dv3std** = genesis_reward_scale 1 (symlog/two-hot as published), return_clamp 0, train_ratio 512, time_limit 400, demos = matched_v2/r2d/dH (terminal 1.0, grant_slack 0) — the ONLY arm eligible for the matrix (A15); genesis_dv3clamp (clamp 100, diagnostic); genesis_dv3fix (clamp + train_ratio 512 + TL 400, diagnostic). std s20: first picks ever (train success 0.22 @284k) with value bounded (~29); clamp s20: 0.25 with value pinned at 98. 3M std pack (s23–25, cluster/sbatch_dv3_pack.sh, 3 runs/GPU, 375k updates ≈ r2dreamer's 373k) with an afterany resume chain (dreamer.py re-enters latest.pt). Disclosures: (i) every dv3 job before 08-28 21:30 ran WITHOUT its overlay (launcher omitted \${EXTRA_CONFIGS} on the real train/eval lines) — the 5M baselines were unaffected; (ii) two other launcher defects (env python, relative demo dir) killed the 08-28 morning batch at start; (iii) models.py clamp is flag-gated (return_clamp, default 0). Eval uses actor.sample() labelled deterministic (mislabel, measurement only). Speed: ~47k sim steps/h (train_ratio 512) unpacked.

5.5 SACfD (legacy)

baselines/rl/train_sacfd_full.py — demo-seeded SB3 SAC (no PER, no n-step, FIFO eviction, staged reward, 400k steps). Superseded by RLPD; retained only as history (absolute-action fling artefact; hardened predicate) (train_rlpd.py:1-22; AUDIT_sources_2026-08-23.md:374; METHODOLOGY.md §6.2).

5.6 Cross-learner invariants

All learners: delta_joint, cap 0.025, leash 0.125, repeat 4, sparse terminal +1 (×100 for WMs), tip rule, same tapes (per world), same IC file, K=5 checkpoints, selection on sel, headline on hold+rnd, 1200-step eval horizon. Differences that remain (disclosure, PREREG §10): state vs pixel modality; per-learner budget and unit (DP 100k grad steps; RLPD 100k decisions; r2d 3e6 sim steps; dv3 1e6); r2d training horizon 400 vs 1200 for the others; r2d/dv3 have replay critics that regress through recorded fail states; recipes tuned on dH; RLPD has no passing positive control.

6. Evaluation protocol

6.1 Initial conditions — baselines/eval_ics.json (generated once by baselines/make_eval_ics.py, seed 0, git dd58c15)

s		
e	content	pointer
t		

sel

demo uids **232 234 235 236 237 239**
242 243 244 245 246 247 248 250 251
(= the historic first-15 demo ICs) —
selection only

eval_ics.js
n;
make_eval_ic
s.py:46,91

hold

252 254 256 265 273 276 284 294 295
302 311 325 327 331 335, drawn
default_rng(0) from the 46 remaining
success uids (61-uid universe), disjoint
from sel — **confirmation split of demo**
ICs, not held-out (11/15 are training ICs
of a 51-demo set; for RLPD every sel/hold
uid is a training reset IC; A8)

make_eval_ic
s.py:55-
58,94-97;
PREREG:277-
280;
AUDIT_prelau
nch:114-118

rnd

support ICs: can xy ~ Uniform over the
solved-demo bounding box ± 1 cm, lo
(0.3395, -0.2370), hi (0.592, 0.1887),
upright, z 0.113; fixed goal; the only novel
ICs

ic_sampling.
py:12-34;
eval_ics.js
n
support_box

Can

uid 234 (sel[1]) is a lying-can IC: picked
0/430 policy-evals across 22 DP runs → sel
max = 14/15 = 0.933; hold has ~1 count of
headroom; BC source claims therefore read
on rnd (N11). N11's "tip rule fires at
decision 1" wording describes the training
env, not the eval env (no tip termination
there) **[CHECK 24]**

PAPER_NOTES.
md:250-255;
AUDIT_INDEX.
md:51

6.2 Harness (cluster/eval_sweep.sh → baselines/wandb_eval.py; r2d via eval_genesis.py/r2d_rescore.sh)

- One episode per fresh process (--ic-file --ic-set --ic-index k --seed k --no-wandb --max-steps 1200), ≤ 1 world per 2 cores (-n 8 → 4 concurrent), resume-safe; missing episodes reported as MISSING, never 0; denominators asserted from the IC file; SWEEP-RESULT line is authoritative (eval_sweep.sh:37,60-66,114-166).
- Sim always CPU (GenesisCanEnv(backend='cpu', max_steps=1200)); DP policy on GPU when visible, SAC on CPU (A4); sim_variant from CLI > sidecar > base, mismatch refuses; action mode/repeat/delta_ref from the checkpoint sidecar (no silent defaults) (wandb_eval.py:122-152,171-210,281-326).
- **Pick metric at eval** = the hardened predicate of §3.5 computed by GenesisCanEnv (no settle); episodes run the full 1200 steps (done = t ≥

max_steps only — **no tip termination at eval**); placed, contact, nested (100-step settle) also recorded but not used for pick-scope headlines (genesis_can_env.py:255-294; PREREG §10).

- Action selection: RLPD deterministic; DP sampled with per-episode seed; r2d sampled (mode reported alongside); dv3 deterministic — recorded in the result JSON (act_selection) with node class, git, sim_variant, ckpt step, n_steps, per-episode records (wandb_eval.py:436-449).
- r2d selection evals run 15 episodes in one process on sel; the protocol readout is the separate fresh-process re-score (§5.3).

6.3 Selection and headline rule (PREREG §5, A7)

K=5 checkpoints at 20/40/60/80/100 % of budget; selection = best sel (ties → later for DP/RLPD); **headline per seed = the selected checkpoint on hold (15) + rnd (30)**, reported by name ("sel / hold / rnd"), plus final-checkpoint hold/rnd and time-to-first-training-pick. Results are quoted only from analysis/results_table.py output (SWEEP-HEADLINE, HEADLINE.txt, RESCORE-RESULT; R2D-RESULT rows are flagged sel-only), deduplicated on (arm, seed), smoke seed 0 excluded, world inferred from wave/seed block (results_table.py:10-40,64-74).

6.4 Seeds per cell (registry mirror as of 08-28; RESULTS_TABLE_2026-08-25.md shows the older n=4/5 snapshot [CHECK 25])

learner	world	reward	arms	n
DP	old	–	dH, dDP, dR2D	5 each (10–14)
DP	corrected	–	dH, dDP	10 each (20–29)
DP	corrected	–	dHallpruned_1e3/1e2, dDPallpruned_1e3/1e2 (N7)	3 planned (1–3 landed)
DP	corrected	–	dH_A/B, dDP_A/B (N14)	3 planned (2/12 landed)

				at audit time; resubmitted 08-28)
RLPD	old	s p a r s e	dH, dDP, dR2D, dDPfails, dR2DDPfails	4 each (10-13)
RLPD	old	d e n s e	dH, dDP, dR2D	4 each
RLPD	cor rec ted	s p a r s e, d e n s e	dH, dDP	10 each (20-29)
r2dre amer (RES COR E)	old	d e n s e	dR2D / dR2DDPfails	4 / 3 at audit (80-83 / 80-82; more re- scores in flight; s83 fails lost)
r2dre amer (sel- only)	old	d e n s e	dH, dDP	3 / 4 (not reporta ble)
dv3	—	—	—	0 at the

6.5 Statistics (PREREG §7 as amended by the 08-28 audit)

- Unit = seed; primary statistic = per-seed success count on `hold` and on `rnd` of the selected checkpoint; paired by seed index within a learner; no pooling of episodes across seeds (episode-pooled Fisher tests are invalid and must not be quoted).
- Tests: Welch t on seed means with **Welch-Satterthwaite df** (N15: hold df 4.52 → CI [-0.259, +0.726]; rnd df 4.17 → [-0.228, +0.556]); exact two-sided permutation test (N15 p = 0.286; **minimum attainable p at 4 v 3 = 2/35 = 0.057**, at 3 v 3 = 2/20 = 0.100); 95 % bootstrap CI on the mean difference as effect size; hierarchical Beta-Binomial P(A>B) as the reported posterior; two primary contrasts per learner → Holm; everything else BH q = 0.1, labelled exploratory (PREREG:172-175,194-201; AUDIT_results §1).
- Run-identity rule A9: a seed id names one run; a rerun replaces; earlier readouts dropped and listed as lost.
- **[CHECK 26]** the only statistics code in `analysis/` is `bayes_source_effect.py` (08-08 wave, `wandb eval_indist/eval_random` keys) and `results_table.py` (means only); the Welch/permutation/Holm computations quoted in the notes were done ad hoc and are not scripted.

UPDATE 08-29. `analysis/stats.py` (Welch with Welch-Satterthwaite df and the correct critical value, exact permutation p under both conventions, minimum attainable p; self-test reproduces the corrected N15 CIs) resolves **[CHECK 26]**. RLPD statistic per A16; WM tables per A19 (one world per table); fails effect per A17 reading rule.

7. Compute, provenance, incidents

item	value	pointer
Cluster	Tufts HPC (Slurm); GPUs L40S / A100 / L40 / H200; one GPU per training run; Genesis worlds CPU-side; partitions <code>gpu, preempt --requeue for training, gpu for relaunches, batch for CPU re-scores (-c 8 --mem 24g --time 4h)</code> ; per-user QOS cap 10 GPUs ; long jobs submitted with <code>--time 24:00:00</code> (no documented 24 h hard cap	<code>sbatch_*.sh</code> headers; <code>sbatch_r2dreamer_pa</code> <code>ck.sh:5-6</code> ; <code>SESSION_LOG:165,173,178</code>

[CHECK 27])

Environments	one pip-only conda env C/condaenv/genesis for genesis + lerobot + SB3 (verified by cluster/verify_env.sh, incl. H200); r2dreader venv C/r2d_venv (py3.11, torch 2.8)	verify_env.sh:24-58; install_r2dreader.sh
Dev box	Ryzen 5950X / RTX 3080Ti / 62 GB; pilots and labs only; official numbers never mixed across machines	METHODOLOGY.md:380-383
Run registry	cluster/run_registry.py → cluster/RUN_REGISTRY.jsonl (cluster copy authoritative, 245 rows 08-18→08-28 in the mirror; local file is empty [CHECK 28]). Row fields: script, arm, seed, knobs, demo_fingerprint (sha of sorted (filename,size)), git, semantic_key, full_key, timestamp, slurm_job_id, duplicate_ok_reason, stage, node; registered at job START; full-key duplicate → REGISTRY-REFUSE unless DUPLICATE_OK=<reason>; semantic duplicate → REGISTRY-WARN (no warn field written yet, contrary to A9 [CHECK 29])	run_registry.py:12-18,57-67,92-102,156-216; sbatch_rlpd.sh:203-235; sbatch_dp.sh:352-381
Registry knobs (RLPD)	steps, budget_unit, scope, action_mode, delta_ref, action_repeat, train/eval horizon, gamma, backup_entropy, per_member_ln, pick_hold_reward, utd, ensemble_size, subset_size, demo_batch, reward, pick_shaping, demo_format, demo_sha, wave, demo_shaping, terminal_zero,	sbatch_rlpd.sh:203-208

	terminal_guard, sim_variant	
Session command log	paper/SESSION_LOG_2026-08-23_cluster.md: every create/run/submit/cancel/delete cluster command with EDT time, host, repo git sha at run time, exact command, exit code, first output line (146 rows + 21 NOTE bullets)	SESSION_LOG:42-43
Artifact mirror	~/workspace/ final_rr_artifacts_2026-08-24/ (9.7 GB): demo recordings + manifests, matched_v2 (incl. r2d/, dR2DDPsucc), matched_w3, DP selected checkpoints and sidecars, RLPD checkpoints (+meta), all .out files (696), HEADLINE.txt (60), sweep.json (577), r2d re-score logs, registry, pull.log	ls M/; AUDIT_INDEX.md §6
Committed provenance	paper/final_rr_sets_2026-08-23/ (v1/v2 manifests + censuses); cluster/patches/ (genesis pin + patch, r2dreamer and dv3 patches); cluster/RUN_REGISTRY.jsonl (should mirror the cluster copy)	
wandb	entity jambotime; projects genesis_paper (DP/RLPD train + sweep summaries, r2dreamer_genesis, dreamer_v3, genesis_pickaplace (wandb_eval default)	eval_sweep.sh:171; sbatch_dp.sh:189; sbatch_rlpd.sh:200; memory monitoring-box-setup.md
Code identity	PREREG §2 "one commit hash per block; a fix restarts that learner's block" — the block spanned several shas (e.g. v2 sets ca05472, w3 sets 2061f08, later relaunches 0440a7f/bf09060); every	SESSION_LOG sha column

registry row and manifest
carries its sha, so per-run
identity is exact even though
the block-level rule was not
kept [**CHECK 30**]

Incidents affecting data (all logged; report in the paper's limitations):

1. **Disk incident (08-27)**: K=5 archiving shipped with no retention → `baselines/outputs` = 909 GB, shared volume 100 %; jobs died with bare exit codes (the same signature previously attributed to preemption/CUDA transients — that diagnosis is partly invalid); 176 checkpoint dirs pruned keeping selected + final of every completed run; 489 GB recovered; of the 20 DP resubmits of 08-27 02:13, 8 completed and 12 failed on the full volume; N14 had 2/12 and N7 9/12 landed before resubmission 08-28 (`SESSION_LOG` NOTE 08-27; `AUDIT_results` §0,§6; commit 30195ee).
2. **Seed overwrite (A9)**: the 08-27 20:15 relaunch reused seed ids of completed runs — `dR2DDP` fails **s83** (sel 0.80, BEST written) overwritten in place and unrecoverable; `dH s102` in-place relaunch cancelled 08-28 04:30 and resubmitted as **s108**; `dDP s101` relaunch never ran. Pooled statistics drop the lost readouts (`PREREG` A9; `SESSION_LOG`:178-179).
3. **Over-broad scancel --name=r2d-train --state=PENDING (08-26)** killed the N13 and N12 top-up waves; 11 relaunched with `DUPLICATE_OK=relaunch-after-overbroad-scancel` (`SESSION_LOG`:162-164).
4. **Re-score false success**: an inline re-score loop printed DONE over 540 crashed processes (no exit-code check; missing `GENESIS_PICKAPLACE_ROOT`); replaced by `cluster/r2d_rescore.sh` with asserted denominators (`r2d_rescore.sh`:11-15; commit bf09060).
5. **E3 exposure arm cancelled** (`BUFFER_MAX=40000` confounded staleness/capacity/reinject shock; 10.1 % vs 34.2 % demo share) — replaced by the N16 controls (`AUDIT_INDEX.md` §4).
6. **Audit-day duplicate submissions**: 24 DP jobs submitted with a wrong guard path, 11 duplicates cancelled before start (`AUDIT_results` §6).
7. Earlier instrument failures that shaped the protocol (fling exploit, env-vs-replay #26 horizon bug, unnormalised dv3 demos, eval action-mode default, six silent-default bugs): `paper/METHODOLOGY.md` §8.4; `CLAUDE.md`.

UPDATE 08-29. GPU packing (CPU-bound Genesis, 0 % GPU util): `r2dreamer` 2-4 runs/GPU, `dv3` 3 runs/GPU, `RLPD` unpacked (GPU-bound at UTD 10). QOS cap 10 GPUs; `gpu MaxTime 2 d`; `scontrol top` denied → queue order via `scontrol hold + --` begin release jobs. Unattended readouts: `cluster/harvest_readout.sh` on the batch partition at +3/+20/+44/+62 h → `paper/harvest_<ts>.md`. Incidents 08-28/29: three silent pack deaths (40 s, found ~8 h later) → rule: check every pack `.out` for `rc=1` within the first hour. Fallback WM (A18): DEMO3 codebase (TD-MPC2-based; MoDem / TD-MPC2 by flags) prepared locally (`~/workspace/demo3_prep`, converter round-trip verified); `policy_pretraining=false` flag set is actor-BC-free.

8. [CHECK] list (every uncertain or conflicting value)

1. **Recording rate:** "30 Hz" (docs) vs ~31.6 Hz (uid 232 `_episodes.npy`, 919 frames/29.06 s) vs ~40 Hz `joint_states` / 1159 frames in `_cartesian.npy` for the same trial; timestamps not saved (`example.py`:276; `CAN_STARTING_POSITION.md`:188).
2. **vel_cmd semantics:** the stored arm stream is measured `/joint_states` position, not a velocity or a command; several docs say "commanded joint targets". Fix wording throughout (`trial_reader.py`:69-74; `METHODOLOGY.md`:233-235,281-284).
3. **Real camera resolution/model:** not documented anywhere.
4. **93 vs 96 trials:** uids 253, 285, 289 (fail list) on disk but absent from `trial_placements.json/manifests`, reason undocumented; also `kinova.py` lists 322 and 331 as both success and fail; "All 224 bags are local" (`HANDOFF_PLAN.md`:87) vs 96 processed.
5. "~130 recordings" for unlabeled uids 200-231 (`HANDOFF_PLAN.md`:91) — inconsistent with a 32-wide range.
6. **Contact-solver default:** `replay_harness.py`:98-99 comment claims 0.2.1 hard-wires `timeconst = 2*substep_dt`; gripper lab shows this tree keeps 0.02 s and only clamps. Confirm which statement holds for the exact pinned commit 31951c3f.
7. **Finger kp:** `CAN_STARTING_POSITION.md`:172-174 says kp 100 "sweet spot"; live world config is `finger_kp 40` (commit e8a07d6).
8. `sim_variants.post_build` does not assert the 5-geom count for `grasp_timeconst` (silent degrade possible while stamping `_ts5`).
9. **Human pick-recreation figures** come from different runs: 51/66 (meas, cluster), 56/66 (either, cluster, base), 57/66 (gripper-lab `g_base` control) vs 58/66 (`ts5`, cluster-confirmed), 58/66 (`dH_w2` in `gc_kp4_riser3_shelf6`), 55/66 and 58-62/66 (local follower-lab variants). State which comparison each number belongs to.
10. **ts5 / matched_w4 fate:** `dH_w4` recorded 08-26 07:50 (`ts5`) and `matched_w4_pilot/dH` built, but no kept count, census, or DP-pilot readout appears in any note or results table; `matched_w4` proper never built; commit 7c8195d calls `ts5` "world of record from 08-26" while all subsequent result launches used `gc_kp4_riser3_shelf6`. The draft states "`ts5` = pilot only, no result block".
11. `genesis_can_env.py`:10 docstring says 16-dim state; code is 17.
12. **Tip penalty:** code `TIP_PENALTY = 0.0` (pick/full scope) and PREREG §2 `r=0`; the task brief's "-0.5 in training" is the July Cartesian env only.
13. **Fast-pick 2.0 mechanism:** the "picked + placed grants in the same window" explanation is read from `genesis_can_env.py`:265-267; the only written source is a comment in `to_dreamer_native.py`:65-70 ("stage grant AND the hardened-pick terminal inside ONE repeat-4 window").
14. `dHunpruned` control was recorded twice but no result for it exists in any results file (PREREG §1 lists `dHunpruned_DP ×3`).
15. PREREG A1's drop taxonomy ("5 over-horizon, 7 over-cap") vs recorder outcome codes (10 `env_truncated`, 2 `adapter_exhausted`, 3 `tipped`) for the same 15 drops.

16. `sbatch_dp.sh:14,221` says lerobot datasets are "prebuilt by `make_matched_sets.py`"; they were built by `convert_to_lerobot.py` in separate steps.
17. Split-half set shas not captured locally.
18. The "22 %" pruned fraction at $\text{eps } 1\text{e-}2$ exists in no artifact.
19. DP 300k "epoch-matched view" (PREREG §1) never ran.
20. RLPD old-world seeds are 10-13 ($n=4$), not 10-14; DP old-world seeds are 10-14 ($n=5$).
21. r2dreamer ran dense only, TL 400, old world — PREREG §1 planned sparse+dense $\times 6$ seeds at TL 1200 (pilot-gated); PREREG §8's on-fail action ("keep 400") was taken but no amendment was logged (CRITIQUE Q6).
22. r2dreamer checkpoint tie-break is `sort -rn | head -1` (not "later"); its in-job selection evals are 15 episodes in one process (not fresh-process).
23. dv3 final-RR block never ran; the N4 negative rests on 08-19 300k runs whose harness (`sbatch_genesis_multi.sh`) lives outside this repo; `results_table.py` has no DV3 parser.
24. N11's "tip rule fires at decision 1" mechanism for uid 234: the eval env has no tip termination; the 0/430 is real but the wording describes the training env.
25. `RESULTS_TABLE_2026-08-25.md` shows $n=4/5$ for corrected-world cells; the mirror/audit have $n=10$. Regenerate before quoting.
26. No scripted Welch/permutation/Holm/BH analysis exists in `analysis/`; `bayes_source_effect.py/results_significance.md` describe the 08-08 wave.
27. No document states a 24 h cluster wall-clock cap; only `--time 24:00:00` usage is observable (script defaults 12 h / 14 h / 2 d).
28. Local `cluster/RUN_REGISTRY.jsonl` is 0 lines; the mirror copy (245 rows) predates seeds 104-108, 200-203, 300-303.
29. Registry rows carry no `warn` field (A9 requires one); 0/245 rows have it.
30. PREREG §2 "one commit hash per block" was not kept (several shas within each block); per-run shas are exact.
31. E5: the ts5 change also stiffens `can↔goal-can` and `can↔shelf` contacts $1.6\times$ — relevant only if any downstream (contact/nested) number from the gripper lab is quoted.
32. `GenesisCanEnv.placed` still tests the un-shifted `BOX_TOP_Z = 0.11` under shelf6 worlds (a correctly placed can scores False); inert for pick scope (CRITIQUE_decisions:455-461).